

Name: _____ Teacher: _____

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS

March 2014

Time Allowed – 70 minutes

DIRECTION TO CANDIDATES:

- All questions may be attempted.
- All questions are not of equal value. The marks indicated are only a guide and may be changed.
- Full marks may not be awarded for careless or badly arranged work, including illegible writing.
- Approved calculators may be used.
- Diagrams are not drawn to scale.
- All necessary working should be shown in every question.
- **Each question attempted is to be started ON A NEW PAGE, clearly marked with the number of the question and your name on the top right hand side of the page.**

SECTION 1

1. Given that the curve $y = ax^2 - 8x - 8$ has a stationary point at $x = 2$, find the value of a
A. $a = \frac{1}{2}$ B. $a = 2$ C. $a = 6$ D. $a = -2$

2. An infinite geometric series has a first term of 8 and a limiting sum of 12. What is the common ratio?
A. $\frac{1}{6}$ B. $\frac{1}{4}$ C. $\frac{1}{3}$ D. $\frac{1}{2}$

3. The equation of the directrix of the parabola $y^2 = -8x$ is
A. $x = 2$ B. $y = 2$ C. $x = -2$ D. $y = -2$

4. $2x + 5$, $3x$ and m form a geometric sequence with a common ratio of 4. The value of m is
A. -4 B. -12 C. -16 D. -48

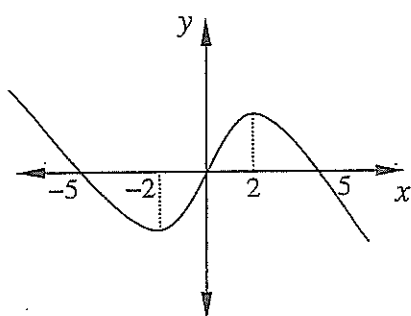
5.

Consider a curve with the following properties:

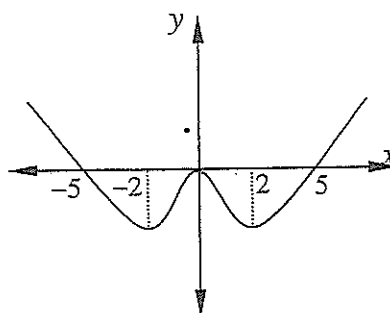
$$\begin{aligned} g(x) &\text{ is odd.} \\ g(5) &= 0 \text{ and } g'(2) = 0. \\ g'(x) &> 0 \text{ for } x > 2. \end{aligned}$$

Which of the following could be the graph of $y = g(x)$?

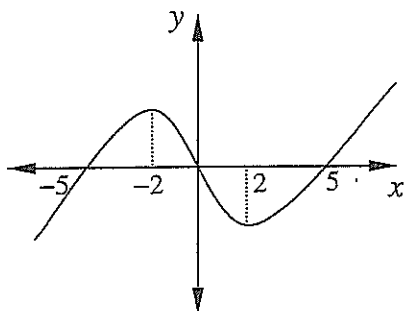
(A)



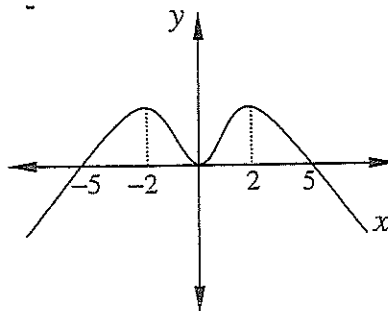
(B)



(C)



(D)



SECTION 2

Question 6 (12 Marks) Start a New page

(a). Find the value of $\sum_{k=0}^5 (k^2 + 1)$ (1)

(b). If α and β are the roots of the equation

$$2x^2 - 6x - 3 = 0, \text{ find the value of}$$

(i) $2\alpha\beta$ (1)

(ii) $(\alpha + \beta)^2$ (1)

(iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ (1)

(iv) $\alpha^2 + \beta^2$ (1)

(c). For the parabola $4y = x^2 + 4x + 12$

(i) Find the co-ordinates of the vertex (2)

(ii) Find the focal length (1)

(iii) Sketch the parabola showing the vertex, focus and directrix (2)

(d) The second term of a geometric series is $\frac{3}{8}$ and the seventh term is 12.

Find the 14th term. (2)

Question 7 (12 marks) Start a New page

(a) The first 3 terms of an arithmetic series are 62, 56 and 50.

(i) Write down a formula for the n th term (1)

(ii) If the last term is -88, how many terms are there in the series? (2)

(iii) Find the sum of the series. (2)

(b) Find the value of k in the equation

$x^2 - (k + 3)x + (k + 6) = 0$ if it has no real roots. (2)

(c) Find the equation of the locus of a point $P(x, y)$ which moves so that line PA is perpendicular to the line PB where $A = (1, 5)$ and $B = (-2, -3)$ (3)

(d) Solve the equation $3^{2x} + 2 \cdot 3^x - 15 = 0$ (2)

Question 8 (12 marks) Start a New page

(a) Consider the curve given by $y = -x^3 + 6x^2 - 9x - 1$

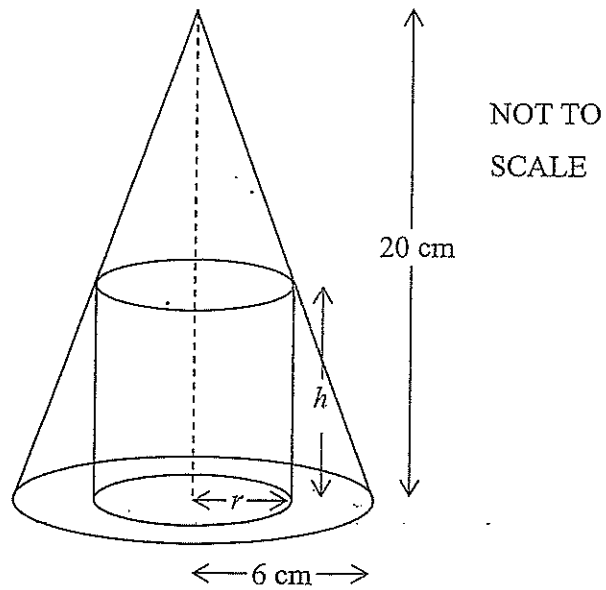
(i) Find the co-ordinates of any stationary points and determine their nature (3)

(ii) Prove a point of inflexion exists and find its co-ordinates (2)

(iii) Sketch the curve for $x \geq 0$, clearly indicating all significant points. (2)

Question 8 Continued

(b)



A cylinder of radius r cm and height h cm is inscribed in a cone with base radius 6 cm and height 20 cm as in the diagram.

(i) Show, using similar triangles, that $h = \frac{10(6-r)}{3}$ (1)

(ii) Show that the volume of the cylinder is given by (1)

$$V = \frac{10\pi r^2(6-r)}{3}$$

(iii) Hence find the values of r and h for the cylinder which has maximum value (3)

Question 9 (14 marks) Start a New Page

(a) Find the primitives of

(i) $8x + 3x^2 - 4x^3$

(ii) $(2x - 1)^3$

(b) Find the equation of the curve passing through the point (2,5) with gradient function $f'(x) = 3x^2 - 4x + 1$.

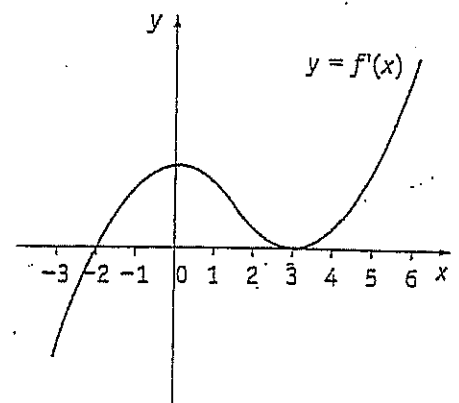
(c) The diagram shows the derivative of $y=f(x)$.

(i) Write down the x co-ordinate of the turning point on $y=f(x)$ and state whether it is a maximum or minimum turning point. (2)

(ii) At what x value on $y=f(x)$ is there a horizontal point of inflexion? (1)

(iii) Where is the function $y=f(x)$ increasing? (1)

(iv) Sketch a possible graph of $y=f(x)$ (1)



(d) On their son Geoffrey's 11th birthday, Mr and Mrs Shum deposited \$600 into an account earning 5% p.a. interest compounded annually. They will continue to deposit \$600 on each of his successive birthdays, up to and including his 21st, giving him the accumulated funds as a present on his 21st birthday.

(i) Show that the amount of Geoffrey's 21st birthday present was \$8524 (to the nearest dollar) (3)

(ii) What single deposit on Geoffrey's 11th birthday would have, under the same conditions, provided the same 21st birthday present? (2)

(d) $3^{2x} + 2 \cdot 3^x - 15 = 0$
 Let $y = 3^x$
 $y^2 + 2y - 15 = 0$
 $(y+5)(y-3) = 0$
 $y = -5$ $y = 3$
 $3^x = -5$ $3^x = 3$
 No SOLN $x = 1$

$$(iv) y = -x^3 + 6x^2 - 9x - 1$$

$$(i) y' = -3x^2 + 12x - 9$$

$$= -3(x^2 - 4x + 3)$$

$$= -3(x-3)(x-1)$$

For st pts let $y' = 0$

$$x = 3, 1$$

$$y = -1, -5$$

$$(3, -1) \quad (1, -5)$$

$$y'' = -6x + 12$$

When $x = 3$ $y'' = -6 < 0$ Max
(3, -1)

When $x = 1$ $y'' = 6 > 0$ Min
(1, -5)

$$(ii) y'' = -6x + 12$$

For pt of inflexion

$$y'' = -6x + 12 = 0$$

$$-6x = -12$$

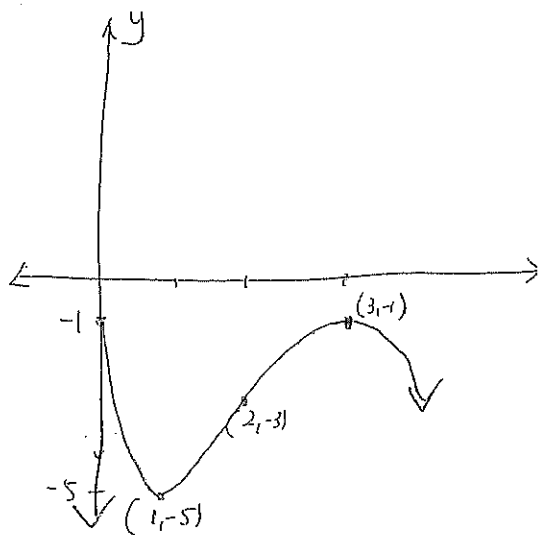
$$x = 2$$

$$y = 3.$$

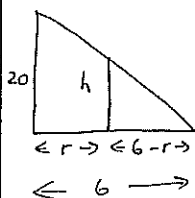
x	1	2	3
y''	6	0	-6

(2, 3) is a pt of inflexion

(iii)



$$(b)(i) V = \pi r^2 h$$



From similar Δ 's

$$\frac{h}{20} = \frac{6-r}{6}$$

$$h = \frac{20(6-r)}{6}$$

$$h = \frac{10(6-r)}{3}$$

$$\therefore V = \pi r^2 \times \frac{10(6-r)}{3}$$

$$V = \frac{10\pi r^2(6-r)}{3}$$

$$(ii) \frac{dV}{dr} = \frac{10\pi}{3} \times \frac{d}{dr}(6r^2 - r^3)$$

$$= \frac{10\pi}{3} \times 12r - 3r^2$$

$$= \frac{10\pi}{3} \times 3r(4-r)$$

$$= 10\pi r(4-r)$$

$$\frac{dV}{dr} = 0 \text{ when } r = 0 \text{ and } r = 4$$

$$\frac{dV}{dr} = 40\pi r - 10\pi r^2$$

$$\frac{d^2V}{dr^2} = 40\pi - 20\pi r$$

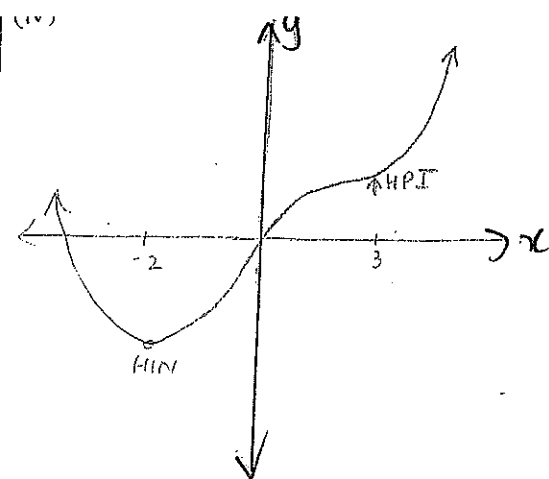
when $r = 4$ $\frac{d^2V}{dr^2} < 0$

$\therefore$ Max when $r = 4$

and $h = \frac{10(6-4)}{3}$

$$h = \frac{20}{3}$$

11th bday $A_1 = 600 \times 1.05^{10}$
 12th " $A_2 = 600 \times 1.05^9$
 :
 20th " $A_{10} = 600 \times 1.05^1$
 21st " $A_{11} = 600$



total $= 600(1.05^{10} + 1.05^9 + \dots + 1)$
 $= 600(1 + 1.05 + \dots + 1.05^{10})$
 $a = 1, n = 11, r = 1.05$

$$= 600 \times \frac{(1.05^{11} - 1)}{0.05}$$

$$= \$8524$$

1) $8524 = x \times 1.05^{10}$

$$x = \$5233$$

2) a.

(i) $x = -2$

minimum

(ii) $x = 3$

(iii) $-2 < x < 3$ and $x > 3$

(c) (i) $\frac{8x^2}{2} + \frac{3x^3}{3} - \frac{4x^4}{4} + C$
 $= 4x^2 + x^3 - x^4 + C$

(ii) $\frac{(2x-1)^4}{2.4} + C$
 $= \frac{(2x-1)^4}{8} + C$

(d) $f'(x) = 3x^2 - 4x + 1$

$$f(x) = x^3 - 2x^2 + x + C$$

At $(2, 5)$

$$5 = 8 - 8 + 2 + C$$

$$C = 3$$

$$\therefore f(x) = x^3 - 2x^2 + x + 3$$